

LECTURE 2: INTRODUCTION TO MONOPOLIES

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WEEK 1

MONOPOLIES

- Last lecture we went through the best case scenario - perfect competition
- This lecture, we go through the worst case scenario of imperfect competition - monopolies!

MONOPOLIES

- Monopolies are basically the worst case scenario in economics
- All of supply is from one firm - this gives it the **power** to set the price, rather than be at the mercy of competition
- A tell-tale sign of monopolies in the real world:
 - ↪ Prices are much higher than recommended retail prices
 - ↪ Restricted supply - that is, making more money from selling less!

NVIDIA

Nvidia is an American company that makes a majority of chips used by the AI and machine learning industry



NVIDIA

Nvidia enjoys pretty close to a monopoly for chips used by AI:

- Not because of chip making - most of its chips are made by a third-party TSMC
- But because of the software that runs on it
- Nvidia make a software, CUDA, that interfaces with large AI models and can only be run on Nvidia chips
- Practically this means that Nvidia has a monopoly on selling chips to AI companies

NVIDIA

This monopoly is at odds with techno-optimists view of the world:

- They say that AI will be ubiquitous and an integral part of every computer

NVIDIA

It is actually quite difficult to buy a new GPU:



Best Buy

No stock. Best Buy remains completely sold out of RTX 5090 cards, and there are currently no RTX 5090 PCs available to order either.

[Click here to find the latest deals from Best Buy.](#)

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Micro Center

GPUs in stock at some stores. As a bricks and mortar store, stock of RTX 5090 cards at Micro Center will depend on your local store. You can check in-store stock on the company's website. There are currently a few RTX 5090 cards available to buy at some stores, with the cheapest option being a Zotac card for \$2,369.99.

[Click here to find the latest deals from Micro Center.](#)

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Where to buy the Nvidia GeForce RTX 5090 in Canada



Amazon Canada

No stuck. While there are a few RTX 5090 graphics cards currently listed on Amazon.ca, none are available to purchase right now, even at (typically) inflated prices. [Click here to find the latest deals from Amazon Canada.](#)

NVIDIA

And if you do, you will probably pay much more than recommended retail price

GPU	Best price	Current MSRP	Premium
RTX 5090	\$3,777.00	\$1,999.00	89%
RTX 5080	\$1,399.99	\$999.00	40%
RTX 5070 Ti	\$839.00	\$749.00	12%
RTX 5070	\$609.00	\$549.00	11%

NVIDIA

Despite selling fewer chips than they could, Nvidia profits are doing just fine!



SO WHAT DO WE LEARN FROM NVIDIA

- Nvidia essentially has monopoly power over supplying chips for AI
- We see the prices in the market are far above recommended retail price
- And, despite selling fewer chips

This comes from Nvidia's monopoly power to sell its particular chips, allowing it to raise prices above its marginal cost of production

In a more competitive market, a business generally becomes less profitable when it sells less than it possibly could

SO WHAT DO WE LEARN FROM NVIDIA

This shows an economic **trade-off**. Nvidia raising its prices:

- Means it will sell fewer chips - losing sales but decreasing costs
- But it will make more profit on the chips that it does sell

We are going to explore how to optimally trade-off losing customers against making more profit on those that stick around

LET'S DO A SIMPLE EXAMPLE WITH NO CALCULUS

The monopolist's goal is to maximize its profits by controlling the price

If demand is $Q_D = 100 - P$, and the firm has no costs, its profit is:

$$\pi = P(100 - P)$$

Let's find what happens at a bunch of different prices!

